10 There are only blue counters, yellow counters, green counters and red counters in a bag.
A counter is taken at random from the bag.

The table shows the probabilities of getting a blue counter or a yellow counter or a green counter.

Colour	blue	yellow	green	red
Probability	0.2	0.35	0.4	

(a) Work out the probability of getting a red counter.

$P(red) = 1 - (0.2 + 0.35 + 0.4) = 0.05$

0.05

1 mark
The probabilities add to 1

(1)

(b) What is the least possible number of counters in the bag?
You must give a reason for your answer.

0.05 = 1 ÷ 20, so red counters are one twentieth of the total.

There must be at least 1 red counter, so at least 20 counters.

1 mark
Least possible total: 20
20 works: blue 4, yellow 7, green 8, red 1 — all whole numbers.

1 mark
Explain why a smaller total cannot work

(2)

(Total for Question 10 is 3 marks)

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